

Static Pod in Kubernetes

- A Static Pod in Kubernetes is a pod that is directly managed by the `kubelet` on a node – not by the Kubernetes API server or controllers like Deployments.
- the kubelet automatically creates and manages it.
 - 👉 No `kubectl apply` needed
 - ❌ No scheduler involved
 - ❌ Runs only on that specific node (each node needs its own manifest file)
 - ❌ No ReplicaSet support
 - Used for critical system workloads
 - 👉 if kubelet restarts, Static Pods are recreated automatically from manifest files

♦ Example Static Pod YAML

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-static
  labels:
    app: nginx
spec:
  containers:
  - name: nginx
    image: nginx
```

♦ How Static Pods Work

- You place a Pod YAML in a directory (e.g., 📌 Save this file in :
`/etc/kubernetes/manifests/nginx-static.yaml`)
 - kubelet watches that directory
 - kubelet automatically creates the pod
 - If the pod crashes → kubelet restarts it
 - If you delete the file → pod is deleted

Key Characteristics

 Feature	 Static Pod
 Managed by	 kubelet
 Created via	 Local file (not API server)
 Scheduler	 Not used
 Runs on	 Specific node only
 Auto restart	 Yes (by kubelet)
 Visible in API	 Mirror Pod

♦ What is a Mirror Pod ?

- Even though static pods are local, kubelet creates a mirror pod in the API server .
- 👉 You can see it using: `kubectl get pods`
- 📌 But you cannot edit/delete it using `kubectl`

What happens if you delete a Static Pod using kubectl ?

- 👉 It gets recreated automatically, Because the YAML file still exists on the node
- 👉 Delete the manifest file: `rm /etc/kubernetes/manifests/pod.yaml` , It delete the Static Pod permanently

Can you expose a Static Pod using a Service ?

- Yes — if it has labels

If API server is down, will Static Pods still run ?

- YES, 👉 Because kubelet manages them locally (That's why control plane runs as Static Pods)

🔥 Real-World Use Cases

- In clusters (like `kubeadm`), control plane components run as static pods : API Server , Scheduler , Controller Manager , etcd
 - 👉 Because they don't depend on the API server
 - 👉 Ensures cluster can self-start (bootstrapping)

! Key Truth (Interview Highlight)

- 👉 In Amazon EKS, you DO NOT manage control plane nodes
 - 👉 So you cannot access `/etc/kubernetes/manifests` on control plane
 - ❌ Meaning : You cannot create or modify Static Pods for control plane components in EKS
 - 👉 You can create Static Pods manually on worker nodes
 - ❌ Not supported on: AWS Fargate (no node access), Control plane nodes (fully managed)
 - ✅ Only works on: EC2 worker nodes
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- ⚡ One-line (Interview 🔥)
 - 👉 Static Pod = Kubelet-managed 📄 + No scheduler ❌ + One Node-specific 📌
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⚡ Static Pods in Kubernetes – Rapid Fire Q&A

 Q#	? Question	 Answer
♦ Q1	What is a Static Pod?	👉 A pod managed directly by the kubelet on a node (not by the API server).
♦ Q2	Who creates Static Pods?	👉 The kubelet .
♦ Q3	Where are Static Pod definitions stored?	👉 On the node filesystem (not in etcd).
🧠 Q4	How does kubelet know about Static Pods?	👉 Via the <code>--pod-manifest-path</code> (or <code>staticPodPath</code>) directory.
🧠 Q5	Example path?	👉 <code>/etc/kubernetes/manifests</code>
⚙️ Q6	What happens if a Static Pod crashes?	👉 Kubelet automatically restarts it.
⚙️ Q7	What if you delete a Static Pod manually?	👉 It gets recreated (because manifest still exists).
⚙️ Q8	Can Static Pods be scheduled?	👉 ❌ No – they are bound to a specific node .
🔍 Q9	What is a Mirror Pod?	👉 A representation of Static Pod in API server .

 Q#	? Question	 Answer
 Q10	Why Mirror Pods exist?	 For visibility via kubectl.
 Q11	Can you edit Mirror Pods?	  No – changes must be made in manifest file .
 Q12	Feature: Static Pod vs Normal Pod	 Managed by: kubelet vs API server; Stored in: Node filesystem vs etcd; Scheduler used:  No vs  Yes; Auto-recreated:  Yes vs Depends on controller
 Q13	Where are Static Pods commonly used?	 Control plane components: kube-apiserver, kube-scheduler, kube-controller-manager
 Q14	Why Static Pods for control plane?	 Ensures components run even if API server is down.
 Q15	Example Static Pod YAML?	 apiVersion: v1 kind: Pod metadata: name: static-nginx spec: containers: - name: nginx image: nginx Place this file in /etc/kubernetes/manifests
 Q16	Can Static Pods use ConfigMaps/Secrets?	  Not directly (limited support).
 Q17	Can they be part of Deployment?	  No
 Q18	Can they use service accounts?	 Limited / not recommended.
 Q19	Static Pod not running?	 Check: Manifest path, kubelet logs, YAML syntax
 Q20	Pod not visible in kubectl?	 Mirror pod may not be created.
 Q21	What is kubelet role here?	 Watches manifest directory & manages pods.
 Q22	What happens if kubelet restarts?	 Static Pods are recreated automatically.
 Q23	Why Static Pods used in kubeadm clusters?	 For control plane bootstrapping.
 Q24	What if API server is down?	 Static Pods still run (managed by kubelet).

1 2 3 4 Q#	? Question	💡 Answer
🎯 Q25	How to update Static Pod?	👉 Modify manifest file → kubelet reloads.
🎯 Q26	Can you scale Static Pods?	👉 ❌ No (manual per node only).

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